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## MANAGING DATA USING PERSISTENT STORAGE

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### Abstract

Various embodiments include systems, methods, and non-transitory computer-readable media for managing data using persistent storage. Consistent with these embodiments, a method includes receiving a write request that includes a value and a key; processing the write request, the processing of the write request including generating a message that includes the value; transmitting the message to an external system for storage of the value; generating an index based on a mapping of the key to an offset; and storing the value and the index in a local persistent storage.

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## Background/Summary

CLAIM OF PRIORITY [0001] This Application is a Continuation of U.S. application Ser. No. 18/073,470, filed Dec. 1, 2022, which is hereby incorporated by reference in its entirety.

### TECHNICAL FIELD

[0002] The present disclosure generally relates to managing data using persistent storage, and, more particularly, various embodiments described herein provide for systems, methods, techniques, instruction sequences, and devices that facilitate efficient data access and management.

### BACKGROUND

[0003] In high throughput systems, data needs to be accessed efficiently to avoid high latency on data processing paths and/or to reduce consumption of compute resources that are also allocated for handling core job execution logic.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0004] In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some embodiments are illustrated by way of example, and not limitation, in the figures of the accompanying drawings.

[0005] FIG. 1 is a block diagram showing an example networked environment that includes a data management system, according to various embodiments of the present disclosure.

[0006] FIG. 2 is a block diagram illustrating an example data management system, according to various embodiments of the present disclosure.

[0007] FIG. 3 is a flowchart illustrating an example method for managing data using persistent storage, according to various embodiments of the present disclosure.

[0008] FIG. 4 is a flowchart illustrating an example method for managing data using persistent storage, according to various embodiments of the present disclosure.

[0009] FIG. 5 is a flowchart illustrating an example method for managing data using persistent storage, according to various embodiments of the present disclosure.

[0010] FIG. 6 is a block diagram illustrating a representative software architecture, which may be used in conjunction with various hardware architectures herein described, according to various embodiments of the present disclosure.

[0011] FIG. 7 is a block diagram illustrating components of a machine able to read instructions from a machine storage medium and perform any one or more of the methodologies discussed herein according to various embodiments of the present disclosure.

### DETAILED DESCRIPTION

[0012] The description that follows includes systems, methods, techniques, instruction sequences, and computing machine program products that embody illustrative embodiments of the present disclosure. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of embodiments. It will be evident, however, to one skilled in the art that the present inventive subject matter may be practiced without these specific details.

[0013] Reference in the specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present subject matter. Thus, the appearances of the phrase “in one embodiment” or “in an embodiment” appearing in various places throughout the

specification are not necessarily all referring to the same embodiment.

[0014] For purposes of explanation, specific configurations and details are set forth in order to provide a thorough understanding of the present subject matter. However, it will be apparent to one of ordinary skill in the art that embodiments of the subject matter described may be practiced without the specific details presented herein, or in various combinations, as described herein. Furthermore, well-known features may be omitted or simplified in order not to obscure the described embodiments. Various embodiments may be given throughout this description. These are merely descriptions of specific embodiments. The scope or meaning of the claims is not limited to the embodiments given.

[0015] Because pieces of the data can be missing due to a lack of access, certain data (e.g., response payloads of executing a job, metadata associated with a response) requiring such missing pieces needs to be persisted before being processed or carried to the next stage of execution. Systems may provide solutions to handle such a situation by generating subgraphs that indicate the next stages of processing for the data that needs to be persisted. However, this approach may be inefficient and may cause system latency, especially when it comes to handling near real-time processing of a large volume of data in high throughput and multifaceted systems.

[0016] Various examples include systems, methods, and non-transitory computer-readable media for managing data using a persistent storage solution. Specifically, the disclosure is related to generating a multilayer key-value store using external storage units (e.g., Kafka partitions or partitions) that are backed by replicated logs. A data management system creates a compacted view (e.g., a copy) of external storage units in a local key-value store on a local persistent storage after consuming the data associated with write requests. The compact view contains copies of messages stored in the external storage units, so that customers can access data in the persistent local storage without needing to retrieve data from the external storage units. The disclosure is further related to addressing issues where messages may contain mutations (e.g., data changes) that apply to different keys of the key-value store. The data management system provides an indexing mechanism where the indexing mechanism translates keys to a list of messages stored on the local persistent storage that contains the keys, and reconstructs the values of the keys when reading the list of messages and applying the set of data changes to recreate the current value of the keys.

[0017] In various embodiments, upon receiving a write request that includes one or more values and one or more keys, a data management system processes the write request including generating one or more messages based on the one or more values and the one or more keys. The one or more messages represent (or include) the one or more values and the one or more keys included in the write request. A message may include a key field and a value field. In various embodiments, the one or more messages includes the one or more values without including the one or more keys associated with the one or more values.

[0018] The data management system transmits the message to an external system for storage (e.g., storage of the value) and generates one or more indexes based on a mapping of the one or more keys to one or more offsets (or offset values) associated with the one or more messages (or the storage of the one or more values). The external system may include one or more storage units, each of which may be a single log file that includes data written in an append-only fashion. An offset value may be an incremental and immutable number, maintained by the external system. In various embodiments, the external system distributes the storage units across multiple clusters for scalability and high message processing throughput. Each cluster may include one or more servers.

[0019] In various embodiments, an index may be generated based on a mapping of a key to an offset and/or a timestamp associated with the message

[0020] In various embodiments, the data management system associates the one or more messages with the one or more indexes and stores the one or more messages and the one or more indexes in a local persistent storage. In various embodiments, the data management system stores the one or more indexes in a local persistent storage without associating the one or more messages with the

one or more indexes.

[0021] In various embodiments, an offset (or an offset value) associated with a message includes an identifier assigned to a storage unit associated with the external system in which the message is stored. In various embodiments, the offset associated with a message may represent a position of the message stored within the storage unit. A timestamp associated with a message may represent a point of time when the message was received or written to the storage unit.

[0022] In various embodiments, a message may be a key-value message where the message includes a key paired with a value. The key may be used to determine the storage unit to which the message appends. The value may correspond to a payload (e.g., actual data) of the message.

[0023] In various embodiments, upon receiving a read request, the data management system identifies the one or more indexes associated with the one or more messages stored in the local persistent storage based on the one or more keys included in the read request. The data management system processes the read request including retrieving the one or more values associated with the one or more keys based on the one or more indexes associated with the one or more messages.

[0024] In various embodiments, the data management system detects that one or more data changes are associated with a key associated with a message. The data management system generates (or reconstructs) a current value of the key based on the one or more data changes and the index associated with the message (or the value) and processes the read request based on the current value of the key.

[0025] In various embodiments, the data management system determines that the one or more values associated with the one or more messages have been removed from the local persistent storage. The data management system reads the one or more values associated with the one or more messages from the storage unit associated with the external system. Specifically, reading the one or more values includes accessing a mapping of the one or more keys to the one or more offsets associated with the one or more messages and providing the one or more offsets to the external system.

[0026] In various embodiments, the data management system determines a frequency of access to the message. The frequency of access refers to the number of occurrences of accessing the message per unit of time (e.g., any number of seconds, minutes, hours, or days).

[0027] In various embodiments, upon determining that the frequency of access to the message is below a threshold value, the data management system deletes the message and the associated index from the local persistent storage. The threshold value may be determined by the data management system based on factors such as system latency, memory storage space, data processing power, etc.) or be determined by a system administrator or an authorized user. In various embodiments, a caching strategy, such as Least Recently Used (LRU) strategy, is used to delete (or evict) elements (e.g., messages) from the memory space (e.g., persistent storage) to make space for new messages when the memory space is full or nearly full.

[0028] In various embodiments, the data management system calculates a cache size based on a plurality of variables, including without limitation, the average rate of director job execution, the average number of items (e.g., messages) written per job execution, the average size of a plurality of messages, and the average retention of items. The data management system estimates a plurality of storage requirements for the local persistent storage based on the cache size.

[0029] In various embodiments, the data management system applies a formula, such as  $E=R \times N \times S \times T$ . R represents the average rate of director job execution. N represents the average number of items written per job execution. S represents the average size of the items. T represents the average retention of items

[0030] Reference will now be made in detail to embodiments of the present disclosure, examples of which are illustrated in the appended drawings. The present disclosure may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth

herein.

[0031] FIG. 1 is a block diagram showing an example networked environment **100** that includes a data management system **122**, according to various embodiments of the present disclosure. By including the data management system **122**, the networked environment **100** can facilitate efficient data access and management in high throughput and multifaceted networked environment as described herein. As shown, the networked environment **100** includes one or more client devices **102**, a server system **108**, and a network **106** (e.g., including Internet, wide-area-network (WAN), local-area-network (LAN), wireless network, etc.) that are communicatively coupled together. Each client device **102** can host a number of applications, including a client software application **104**. The client software application **104** can communicate data with the server system **108** via a network **106**. Accordingly, the client software application **104** can communicate and exchange data with the server system **108** via the network **106**.

[0032] The server system **108** provides server-side functionality via the network **106** to the client software application **104**. While certain functions of the networked environment **100** are described herein as being performed by the data management system **122** on the server system **108**, it will be appreciated that the location of certain functionality within the server system **108** is a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the server system **108**, but later migrate this technology and functionality to the client software application **104**.

[0033] With respect to the server system **108**, each of an Application Program Interface (API) server **110** and a web server **112** is coupled to an application server **116**, which hosts the data management system **122**. The application server **116** is communicatively coupled to a database server **118**, which facilitates access to a database **120** that stores data associated with the application server **116**, including data that may be generated or used by the data management system **122**, in various embodiments.

[0034] The API server **110** receives and transmits data (e.g., API calls, commands, requests, responses, and authentication data) between the client device **102** and the application server **116**, and between the one or more services and the application server **116**. Specifically, the API server **110** provides a set of interfaces (e.g., endpoint, routines, or protocols) that can be called or queried by the client software application **104** or the one or more services in order to invoke the functionalities of the application server **116**. The API server **110** exposes various functions supported by the application server **116**, including without limitation: user registration; login functionality; data object operations (e.g., generating, storing, retrieving, encrypting, decrypting, transferring, access rights, licensing, etc.), and user communications.

[0035] Through one or more web-based interfaces (e.g., web-based user interfaces), the web server **112** can support various functions of the data management system **122** of the application server **116**, including without limitation: receiving a write request that includes a value and a key; processing the write request including generating a message that represents the value and the key; transmitting the message to an external system (e.g., hosted by the third-party platform **124**) for storage; generating an index based on a mapping of the key to an offset and/or a timestamp associated with the message; associating the message with the index; storing the message and the index in a local persistent storage; receiving a read request that includes the key; identifying, based on the key, the index associated with the message stored in the local persistent storage; and processing the read request including retrieving the value associated with the key based on the index associated with the message (or the storage of the value). In various embodiments, the deployment or implementation of the web server **112** and the application server **116** may share the same set of executable code. In various embodiments, the web server **112** may be a subsystem or a component of the application server **116**.

[0036] The application server **116** hosts a number of applications and subsystems, including the data management system **122**, which supports various functions and services with respect to

various embodiments described herein. The application server **116** is communicatively coupled to a database server **118**, which facilitates access to database(s) **120** in which may be stored data associated with the data management system **122**.

[0037] In various embodiments, database(s) **120** may include one or more persistent storage units (also referred to as persistent storage) that store messages as described herein.

[0038] FIG. **2** is a block diagram illustrating an example data management system **200**, according to various embodiments of the present disclosure. For some embodiments, the data management system **200** represents an example of the data management system **122** described with respect to FIG. **1**. As shown, the data management system **200** comprises a request receiving component **210**, a request processing component **220**, an index generating and associating component **230**, a message storing component **240**, an index identifying component **250**, and a key value retrieving component **260**. According to various embodiments, one or more of the request receiving component **210**, the request processing component **220**, the index generating and associating component **230**, the message storing component **240**, the index identifying component **250**, and the key value retrieving component **260** are implemented by one or more hardware processors **202**. Data generated by one or more of the request receiving component **210**, the request processing component **220**, the index generating and associating component **230**, the message storing component **240**, the index identifying component **250**, and the key value retrieving component **260** is stored in a database **270** of the data management system **200**.

[0039] In various embodiments, the request receiving component **210** is configured to receive one or more write requests that include one or more values and one or more keys. The request receiving component **210** is further configured to receive one or more read requests of one or more messages previously written to a persistent storage and/or a storage unit associated with an external system, such as one hosted by the third-party platform **124**, as illustrated in FIG. **1**.

[0040] In various embodiments, the request processing component **220** is configured to process the one or more write requests. The processing includes generating one or more messages based on the one or more values and the one or more keys. The one or more messages represent (or include) the one or more values and the one or more keys included in the write request. A message may include a key field and a value field. The external system may include one or more storage units. Each storage unit may be a single log file that includes data written in an append-only fashion.

[0041] In various embodiments, the request processing component **220** is further configured to transmit the one or more messages to an external system for storage.

[0042] In various embodiments, the index generating and associating component **230** is configured to generate one or more indexes based on a mapping of the one or more keys to one or more offsets (or offset values) and/or one or more timestamps associated with the one or more messages. The index generating and associating component **230** is further configured to associate the one or more messages with the one or more indexes.

[0043] In various embodiments, the message storing component **240** is configured to store the one or more messages and the one or more indexes in a local persistent storage.

[0044] In various embodiments, upon receiving one or more read requests, the index identifying component **250** is configured to identify the one or more indexes associated with one or more messages (or the storage of the one or more values) stored in the local persistent storage based on one or more keys included in the one or more read request.

[0045] In various embodiments, the key value retrieving component **260** is configured to retrieve one or more values associated with the one or more keys based on the identified one or more indexes associated with the one or more messages.

[0046] FIG. **3** is a flowchart illustrating an example method **300** for managing data using persistent storage, according to various embodiments of the present disclosure. It will be understood that example methods described herein may be performed by a machine in accordance with some embodiments. For example, the methods **300** can be performed by the data management system

**122** described with respect to FIG. **1**, the data management system **200** described with respect to FIG. **2**, or individual components thereof. An operation of various methods described herein may be performed by one or more hardware processors (e.g., central processing units or graphics processing units) of a computing device (e.g., a desktop, server, laptop, mobile phone, tablet, etc.), which may be part of a computing system based on a cloud architecture. Example methods described herein may also be implemented in the form of executable instructions stored on a machine-readable medium or in the form of electronic circuitry. For instance, the operations of method **300** may be represented by executable instructions that, when executed by a processor of a computing device, cause the computing device to perform method **300**. Depending on the embodiment, an operation of an example method described herein may be repeated in different ways or involve intervening operations not shown. Though the operations of example methods may be depicted and described in a certain order, the order in which the operations are performed may vary among embodiments, including performing certain operations in parallel.

[0047] At operation **302**, a processor receives a write request that includes one or more values and one or more keys.

[0048] At operation **304**, a processor processes the write request, including generating one or more messages based on the one or more values and the one or more keys. The one or more messages represent (or include) the one or more values and the one or more keys included in the write request.

[0049] At operation **306**, a processor transmits the one or more messages to an external system for storage. The external system may include one or more storage units.

[0050] At operation **308**, a processor generates one or more indexes based on a mapping of the one or more keys to one or more offsets (or offset values) associated with the one or more messages. In various embodiments, an index may be generated based on a mapping of a key to an offset and/or a timestamp associated with the message. In various embodiments, a processor associates the one or more messages with the one or more indexes and stores.

[0051] At operation **310**, a processor only stores the one or more indexes in a local persistent storage.

[0052] In various embodiments, a processor stores the one or more messages and the one or more indexes in a local persistent storage, on a local disk (e.g., cache), for example. The local persistent storage provides a compacted view (e.g., a copy) of external storage units in a local key-value store on a local persistent storage after consuming the data. The compact view contains copies of messages stored in the external storage units, so that customers can quickly access data from the local persistent storage without having to retrieve data from the external storage units.

[0053] Though not illustrated, the method **300** can include an operation where a graphical user interface for managing data can be displayed (or caused to be displayed) by the hardware processor. For instance, the operation can cause a computing device to display the graphical user interface for managing data. This operation for displaying the graphical user interface can be separate from operations **302** through **310** or, alternatively, form part of one or more of operations **302** through **310**.

[0054] FIG. **4** is a flowchart illustrating an example method **400** for managing data using persistent storage, according to various embodiments of the present disclosure. It will be understood that example methods described herein may be performed by a machine in accordance with some embodiments. For example, the methods **400** can be performed by the data management system **122** described with respect to FIG. **1**, the data management system **200** described with respect to FIG. **2**, or individual components thereof. An operation of various methods described herein may be performed by one or more hardware processors (e.g., central processing units or graphics processing units) of a computing device (e.g., a desktop, server, laptop, mobile phone, tablet, etc.), which may be part of a computing system based on a cloud architecture. Example methods described herein may also be implemented in the form of executable instructions stored on a

machine-readable medium or in the form of electronic circuitry. For instance, the operations of method **400** may be represented by executable instructions that, when executed by a processor of a computing device, cause the computing device to perform method **400**. Depending on the embodiment, an operation of an example method described herein may be repeated in different ways or involve intervening operations not shown. Though the operations of example methods may be depicted and described in a certain order, the order in which the operations are performed may vary among embodiments, including performing certain operations in parallel.

[0055] In various embodiments, one or more operations of the method **400** may be a sub-routine of one or more of the operations of method **300**. In various embodiments, one or more operations in method **400** may be performed subsequent to the operations of method **300**.

[0056] At operation **402**, a processor receives one or more read requests that include one or more keys.

[0057] At operation **404**, a processor identifies one or more indexes associated with the one or more messages stored in the local persistent storage based on the one or more keys included in the one or more read requests.

[0058] In various embodiments, operation **404** is optional where the one or more values stored in the local persistent storage (e.g., a cache) may be retrieved based on the one or more keys. The indexes are only needed when the values are not stored locally that the values are stored only in the external system.

[0059] At operation **406**, a processor processes the one or more read requests, including retrieving the one or more values associated with the one or more keys from a cache in the local persistent storage.

[0060] In various embodiments, a processor processes the one or more read requests, including retrieving one or more values associated with the one or more keys based on the one or more indexes associated with the one or more messages.

[0061] Though not illustrated, the method **400** can include an operation where a graphical user interface for managing data using persistent storage can be displayed (or caused to be displayed) by the hardware processor. For instance, the operation can cause a computing device to display the graphical user interface for managing data using persistent storage. This operation for displaying the graphical user interface can be separate from operations **402** through **406** or, alternatively, form part of one or more of operations **402** through **406**.

[0062] FIG. 5 is a flowchart illustrating an example method **500** for managing data using persistent storage, according to various embodiments of the present disclosure. It will be understood that example methods described herein may be performed by a machine in accordance with some embodiments. For example, the methods **500** can be performed by the data management system **122** described with respect to FIG. 1, the data management system **200** described with respect to FIG. 2, or individual components thereof. An operation of various methods described herein may be performed by one or more hardware processors (e.g., central processing units or graphics processing units) of a computing device (e.g., a desktop, server, laptop, mobile phone, tablet, etc.), which may be part of a computing system based on a cloud architecture. Example methods described herein may also be implemented in the form of executable instructions stored on a machine-readable medium or in the form of electronic circuitry. For instance, the operations of method **500** may be represented by executable instructions that, when executed by a processor of a computing device, cause the computing device to perform method **500**. Depending on the embodiment, an operation of an example method described herein may be repeated in different ways or involve intervening operations not shown. Though the operations of example methods may be depicted and described in a certain order, the order in which the operations are performed may vary among embodiments, including performing certain operations in parallel.

[0063] In various embodiments, one or more operations of the method **500** may be a sub-routine of one or more of the operations of method **300** and/or method **400**. In various embodiments, one or



more operations in method **500** may be performed subsequent to the operations of method **300** and/or method **400**.

[0064] At operation **502**, a processor detects that one or more data changes are associated with a key associated with a message.

[0065] At operation **504**, a processor generates (or reconstructs) a current value of the key based on the one or more data changes. In various embodiments, a processor generates (or reconstructs) a current value of the key based on the one or more data changes and the index associated with the message.

[0066] At operation **506**, a processor causes the current value of the key to be stored in the external system.

[0067] In various embodiments, a processor processes the read request based on the current value of the key. The processing of the read request includes providing the value of the message to the sender of the read request.

[0068] Though not illustrated, the method **500** can include an operation where a graphical user interface for managing data using persistent storage can be displayed (or caused to be displayed) by the hardware processor. For instance, the operation can cause a computing device to display the graphical user interface for managing data using persistent storage. This operation for displaying the graphical user interface can be separate from operations **502** through **506** or, alternatively, form part of one or more of operations **502** through **506**.

[0069] FIG. **6** is a block diagram illustrating an example of a software architecture **602** that may be installed on a machine, according to some example embodiments. FIG. **6** is merely a non-limiting example of software architecture, and it will be appreciated that many other architectures may be implemented to facilitate the functionality described herein. The software architecture **602** may be executing on hardware such as a machine **700** of FIG. **7** that includes, among other things, processors **710**, memory **730**, and input/output (I/O) components **750**. A representative hardware layer **604** is illustrated and can represent, for example, the machine **700** of FIG. **7**. The representative hardware layer **604** comprises one or more processing units **606** having associated executable instructions **608**. The executable instructions **608** represent the executable instructions of the software architecture **602**. The hardware layer **604** also includes memory or storage modules **610**, which also have the executable instructions **608**. The hardware layer **604** may also comprise other hardware **612**, which represents any other hardware of the hardware layer **604**, such as the other hardware illustrated as part of the machine **800**.

[0070] In the example architecture of FIG. **6**, the software architecture **602** may be conceptualized as a stack of layers, where each layer provides particular functionality. For example, the software architecture **602** may include layers such as an operating system **614**, libraries **616**, frameworks/middleware **618**, applications **620**, and a presentation layer **644**. Operationally, the applications **620** or other components within the layers may invoke API calls **624** through the software stack and receive a response, returned values, and so forth (illustrated as messages **626**) in response to the API calls **624**. The layers illustrated are representative in nature, and not all software architectures have all layers. For example, some mobile or special-purpose operating systems may not provide a frameworks/middleware **618** layer, while others may provide such a layer. Other software architectures may include additional or different layers.

[0071] The operating system **614** may manage hardware resources and provide common services. The operating system **614** may include, for example, a kernel **628**, services **630**, and drivers **632**. The kernel **628** may act as an abstraction layer between the hardware and the other software layers. For example, the kernel **628** may be responsible for memory management, processor management (e.g., scheduling), component management, networking, security settings, and so on. The services **630** may provide other common services for the other software layers. The drivers **632** may be responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **632** may include display drivers, camera drivers, Bluetooth® drivers, flash memory drivers, serial

communication drivers (e.g., Universal Serial Bus (USB) drivers), Wi-Fi® drivers, audio drivers, power management drivers, and so forth depending on the hardware configuration.

[0072] The libraries **616** may provide a common infrastructure that may be utilized by the applications **620** and/or other components and/or layers. The libraries **616** typically provide functionality that allows other software modules to perform tasks in an easier fashion than by interfacing directly with the underlying operating system **614** functionality (e.g., kernel **628**, services **630**, or drivers **632**). The libraries **616** may include system libraries **634** (e.g., C standard library) that may provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **616** may include API libraries **636** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as MPEG4, H.264, MP3, AAC, AMR, JPG, and PNG), graphics libraries (e.g., an OpenGL framework that may be used to render 2D and 3D graphic content on a display), database libraries (e.g., SQLite that may provide various relational database functions), web libraries (e.g., WebKit that may provide web browsing functionality), and the like. The libraries **616** may also include a wide variety of other libraries **638** to provide many other APIs to the applications **620** and other software components/modules.

[0073] The frameworks **618** (also sometimes referred to as middleware) may provide a higher-level common infrastructure that may be utilized by the applications **620** or other software components/modules. For example, the frameworks **618** may provide various graphical user interface functions, high-level resource management, high-level location services, and so forth. The frameworks **618** may provide a broad spectrum of other APIs that may be utilized by the applications **620** and/or other software components/modules, some of which may be specific to a particular operating system or platform.

[0074] The applications **620** include built-in applications **640** and/or third-party applications **642**. Examples of representative built-in applications **640** may include, but are not limited to, a home application, a contacts application, a browser application, a book reader application, a location application, a media application, a messaging application, or a game application.

[0075] The third-party applications **642** may include any of the built-in applications **640**, as well as a broad assortment of other applications. In a specific example, the third-party applications **642** (e.g., an application developed using the Android™ or iOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as iOS™, Android™, or other mobile operating systems. In this example, the third-party applications **642** may invoke the API calls **624** provided by the mobile operating system such as the operating system **614** to facilitate functionality described herein.

[0076] The applications **620** may utilize built-in operating system functions (e.g., kernel **628**, services **630**, or drivers **632**), libraries (e.g., system libraries **634**, API libraries **636**, and other libraries **638**), or frameworks/middleware **618** to create user interfaces to interact with users of the system. Alternatively, or additionally, in some systems, interactions with a user may occur through a presentation layer, such as the presentation layer **644**. In these systems, the application/module “logic” can be separated from the aspects of the application/module that interact with the user.

[0077] Some software architectures utilize virtual machines. In the example of FIG. 6, this is illustrated by a virtual machine **648**. The virtual machine **648** creates a software environment where applications/modules can execute as if they were executing on a hardware machine (e.g., the machine **700** of FIG. 7). The virtual machine **648** is hosted by a host operating system (e.g., the operating system **614**) and typically, although not always, has a virtual machine monitor **646**, which manages the operation of the virtual machine **648** as well as the interface with the host operating system (e.g., the operating system **614**). A software architecture executes within the virtual machine **648**, such as an operating system **650**, libraries **652**, frameworks/middleware **654**, applications **656**, or a presentation layer **658**. These layers of software architecture executing within the virtual machine **648** can be the same as corresponding layers previously described or

may be different.

[0078] FIG. 7 illustrates a diagrammatic representation of a machine **700** in the form of a computer system within which a set of instructions may be executed for causing the machine **700** to perform any one or more of the methodologies discussed herein, according to an embodiment. Specifically, FIG. 7 shows a diagrammatic representation of the machine **700** in the example form of a computer system, within which instructions **716** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **700** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **716** may cause the machine **700** to execute the method **300** described above with respect to FIG. 3, the method **400** described above with respect to FIG. 4, and the method **500** described above with respect to FIG. 5. Instructions **716** transform the general, non-programmed machine **700** into a particular machine **700** programmed to carry out the described and illustrated functions in the manner described. In alternative embodiments, the machine **700** operates as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **700** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **700** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smart phone, a mobile device, or any machine capable of executing the instructions **716**, sequentially or otherwise, that specify actions to be taken by the machine **700**. Further, while only a single machine **700** is illustrated, the term “machine” shall also be taken to include a collection of machines **700** that individually or jointly execute the instructions **716** to perform any one or more of the methodologies discussed herein.

[0079] The machine **700** may include processors **710**, memory **730**, and I/O components **750**, which may be configured to communicate with each other such as via a bus **702**. In an embodiment, the processors **710** (e.g., a hardware processor, such as a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a radio-frequency integrated circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **712** and a processor **714** that may execute the instructions **716**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. 7 shows multiple processors **710**, the machine **700** may include a single processor with a single core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

[0080] The memory **730** may include a main memory **732**, a static memory **734**, and a storage unit **736** including machine-readable medium **738**, each accessible to the processors **710** such as via the bus **702**. The main memory **732**, the static memory **734**, and the storage unit **736** store the instructions **716** embodying any one or more of the methodologies or functions described herein. The instructions **716** may also reside, completely or partially, within the main memory **732**, within the static memory **734**, within the storage unit **736**, within at least one of the processors **710** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **700**.

[0081] The I/O components **750** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **750** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones will likely include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O

components **750** may include many other components that are not shown in FIG. 7. The I/O components **750** are grouped according to functionality merely for simplifying the following discussion, and the grouping is in no way limiting. In various embodiments, the I/O components **750** may include output components **752** and input components **754**. The output components **752** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The input components **754** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and/or force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

[0082] In further embodiments, the I/O components **750** may include biometric components **756**, motion components **758**, environmental components **760**, or position components **762**, among a wide array of other components. The motion components **758** may include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The environmental components **760** may include, for example, illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detect concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position components **762** may include location sensor components (e.g., a Global Positioning System (GPS) receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

[0083] Communication may be implemented using a wide variety of technologies. The I/O components **750** may include communication components **764** operable to couple the machine **700** to a network **780** or devices **770** via a coupling **782** and a coupling **772**, respectively. For example, the communication components **764** may include a network interface component or another suitable device to interface with the network **780**. In further examples, the communication components **764** may include wired communication components, wireless communication components, cellular communication components, near field communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **770** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

[0084] Moreover, the communication components **764** may detect identifiers or include components operable to detect identifiers. For example, the communication components **764** may include radio frequency identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **764**, such as location via Internet Protocol (IP) geolocation, location

via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

[0085] Certain embodiments are described herein as including logic or a number of components, modules, elements, or mechanisms. Such modules can constitute either software modules (e.g., code embodied on a machine-readable medium or in a transmission signal) or hardware modules. A “hardware module” is a tangible unit capable of performing certain operations and can be configured or arranged in a certain physical manner. In various example embodiments, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware modules of a computer system (e.g., a processor or a group of processors) are configured by software (e.g., an application or application portion) as a hardware module that operates to perform certain operations as described herein.

[0086] In some embodiments, a hardware module is implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware module can include dedicated circuitry or logic that is permanently configured to perform certain operations. For example, a hardware module can be a special-purpose processor, such as a field-programmable gate array (FPGA) or an ASIC. A hardware module may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware module can include software encompassed within a general-purpose processor or other programmable processor. It will be appreciated that the decision to implement a hardware module mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) can be driven by cost and time considerations.

[0087] Accordingly, the phrase “module” should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering embodiments in which hardware modules are temporarily configured (e.g., programmed), each of the hardware modules need not be configured or instantiated at any one instance in time. For example, where a hardware module comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware modules) at different times. Software can accordingly configure a particular processor or processors, for example, to constitute a particular hardware module at one instance of time and to constitute a different hardware module at a different instance of time.

[0088] Hardware modules can provide information to, and receive information from, other hardware modules. Accordingly, the described hardware modules can be regarded as being communicatively coupled. Where multiple hardware modules exist contemporaneously, communications can be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware modules. In embodiments in which multiple hardware modules are configured or instantiated at different times, communications between or among such hardware modules may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware modules have access. For example, one hardware module performs an operation and stores the output of that operation in a memory device to which it is communicatively coupled. A further hardware module can then, at a later time, access the memory device to retrieve and process the stored output. Hardware modules can also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).

[0089] The various operations of example methods described herein can be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors constitute processor-implemented modules that operate to perform one

or more operations or functions described herein. As used herein, “processor-implemented module” refers to a hardware module implemented using one or more processors.

[0090] Similarly, the methods described herein can be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method can be performed by one or more processors or processor-implemented modules. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines **700** including processors **710**), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). In certain embodiments, for example, a client device may relay or operate in communication with cloud computing systems and may access circuit design information in a cloud environment.

[0091] The performance of certain of the operations may be distributed among the processors, not only residing within a single machine **700**, but deployed across a number of machines **700**. In some example embodiments, the processors **710** or processor-implemented modules are located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the processors or processor-implemented modules are distributed across a number of geographic locations.

#### Executable Instructions and Machine Storage Medium

[0092] The various memories (i.e., **730**, **732**, **734**, and/or the memory of the processor(s) **710**) and/or the storage unit **736** may store one or more sets of instructions **716** and data structures (e.g., software) embodying or utilized by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **716**), when executed by the processor(s) **710**, cause various operations to implement the disclosed embodiments.

[0093] As used herein, the terms “machine-storage medium,” “device-storage medium,” and “computer-storage medium” mean the same thing and may be used interchangeably. The terms refer to a single or multiple storage devices and/or media (e.g., a centralized or distributed database, and/or associated caches and servers) that store executable instructions **716** and/or data. The terms shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and/or device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium” discussed below.

#### Transmission Medium

[0094] In various embodiments, one or more portions of the network **780** may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a LAN, a wireless LAN (WLAN), a WAN, a wireless WAN (WWAN), a metropolitan-area network (MAN), the Internet, a portion of the Internet, a portion of the public switched telephone network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, the network **780** or a portion of the network **780** may include a wireless or cellular network, and the coupling **782** may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or another type of cellular or wireless coupling. In this example, the coupling **782** may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized

(EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High-Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long-Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

[0095] The instructions may be transmitted or received over the network using a transmission medium via a network interface device (e.g., a network interface component included in the communication components) and utilizing any one of a number of well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions may be transmitted or received using a transmission medium via the coupling (e.g., a peer-to-peer coupling) to the devices 770. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure. The terms “transmission medium” and “signal medium” shall be taken to include any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by the machine, and include digital or analog communications signals or other intangible media to facilitate communication of such software. Hence, the terms “transmission medium” and “signal medium” shall be taken to include any form of modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal.

#### Computer-Readable Medium

[0096] The terms “machine-readable medium,” “computer-readable medium,” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure. The terms are defined to include both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. For instance, an embodiment described herein can be implemented using a non-transitory medium (e.g., a non-transitory computer-readable medium).

[0097] Throughout this specification, plural instances may implement resources, components, operations, or structures described as a single instance. Although individual operations of one or more methods are illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in example configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components.

[0098] As used herein, the term “or” may be construed in either an inclusive or exclusive sense. The terms “a” or “an” should be read as meaning “at least one,” “one or more,” or the like. The presence of broadening words and phrases such as “one or more,” “at least,” “but not limited to,” or other like phrases in some instances shall not be read to mean that the narrower case is intended or required in instances where such broadening phrases may be absent. Additionally, boundaries between various resources, operations, modules, engines, and data stores are somewhat arbitrary, and particular operations are illustrated in a context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within a scope of various embodiments of the present disclosure. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense.

[0099] It will be understood that changes and modifications may be made to the disclosed embodiments without departing from the scope of the present disclosure. These and other changes or modifications are intended to be included within the scope of the present disclosure.

## Claims

- 1.** A method comprising: receiving a read request that comprises a key; identifying an index based on the key, the index corresponding to one or more messages stored in a cache of a local persistent storage; reconstructing a value of the key based on the one or more messages; and processing the read request based on the value of the key.
- 2.** The method of claim 1, comprising: determining that the one or more messages comprise data changes to the key, and reconstructing the value of the key comprising applying the data changes to content of the one or more messages.
- 3.** The method of claim 1, comprising: determining that the one or more messages have been removed from the cache of the local persistent storage; and reading the one or more messages from an external system, the reading comprising accessing a mapping of the key to one or more offsets associated with the one or more messages, and providing the one or more offsets to the external system.
- 4.** The method of claim 3, wherein an offset from the one or more offsets comprises an identifier assigned to a storage unit associated with the external system and wherein the offset represents a position of a message stored in the storage unit.
- 5.** The method of claim 3, wherein the local persistent storage provides a compact view of messages stored in the external system in a local key-value store.
- 6.** The method of claim 1, wherein the key is a first key, wherein the value is a first value, wherein the index is a first index, comprising: receiving a write request that comprises a second value and a second key; processing the write request, the processing of the write request comprising generating a message that comprises the second value; transmitting the message to an external system for storage, the message being associated with an offset value indicating a storage position of the message in the external system; generating a second index based on the offset value associated with the message and the second key; and storing the second index in the local persistent storage.
- 7.** The method of claim 6, further comprising: storing the second value in the cache of the local persistent storage; determining that a frequency of access to the second value from the cache in the local persistent storage is below a threshold value; and deleting the second value and the second index from the cache of the local persistent storage.
- 8.** The method of claim 7, wherein the threshold value is determined based on one or more factors comprising system latency, memory storage space, data processing power, and a configuration determined by a system administrator or an authorized user.
- 9.** The method of claim 1, further comprising: accessing a plurality of variables; applying a formula to calculate a cache size based on one or more variables; and estimating a plurality of storage requirements for the local persistent storage based on the cache size.
- 10.** The method of claim 9, wherein the one or more variables comprises one or more of an average rate of director job execution, an average number of items written per job execution, an average size of a plurality of messages, and an average retention of items.
- 11.** A system comprising: at least one memory storing instructions; and one or more hardware processors communicatively coupled to the memory and configured by the instructions to perform operations comprising: receiving a read request that comprises a key; identifying an index based on the key, the index corresponding to one or more messages stored in a cache of a local persistent storage; reconstructing a value of the key based on the one or more messages; and processing the read request based on the value of the key.
- 12.** The system of claim 11, wherein the operations comprise: determining that the one or more messages comprise data changes to the key, and reconstructing the value of the key comprising applying the data changes to content of the one or more messages.
- 13.** The system of claim 11, wherein the operations comprise: determining that the one or more messages have been removed from the cache of the local persistent storage; and reading the one or more messages from an external system, the reading comprising accessing a mapping of the key to



one or more offsets associated with the one or more messages, and providing the one or more offsets to the external system.

**14.** The system of claim 13, wherein an offset from the one or more offsets comprises an identifier assigned to a storage unit associated with the external system and wherein the offset represents a position of a message stored in the storage unit.

**15.** The system of claim 13, wherein the local persistent storage provides a compact view of messages stored in the external system in a local key-value store.

**16.** The system of claim 11, wherein the key is a first key, wherein the value is a first value, wherein the index is a first index, and wherein the operations comprise: receiving a write request that comprises a second value and a second key; processing the write request, the processing of the write request comprising generating a message that comprises the second value; transmitting the message to an external system for storage, the message being associated with an offset value indicating a storage position of the message in the external system; generating a second index based on the offset value associated with the message and the second key; and storing the second index in the local persistent storage.

**17.** The system of claim 16, wherein the operations comprise: storing the second value in the cache of the local persistent storage; determining that a frequency of access to the second value from the cache in the local persistent storage is below a threshold value; and deleting the second value and the second index from the cache of the local persistent storage.

**18.** The system of claim 17, wherein the threshold value is determined based on one or more factors comprising system latency, memory storage space, data processing power, and a configuration determined by a system administrator or an authorized user.

**19.** The system of claim 11, wherein the operations comprise: accessing a plurality of variables; applying a formula to calculate a cache size based on one or more variables; and estimating a plurality of storage requirements for the local persistent storage based on the cache size.

**20.** A non-transitory computer-readable medium comprising instructions that, when executed by a hardware processor of a device, cause the device to perform operations comprising: receiving a read request that comprises a key; identifying an index based on the key, the index corresponding to one or more messages stored in a cache of a local persistent storage; reconstructing a value of the key based on the one or more messages; and processing the read request based on the value of the key.

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